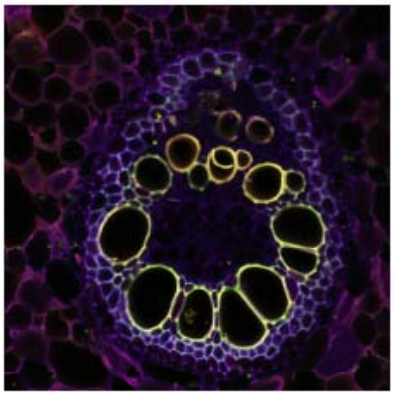
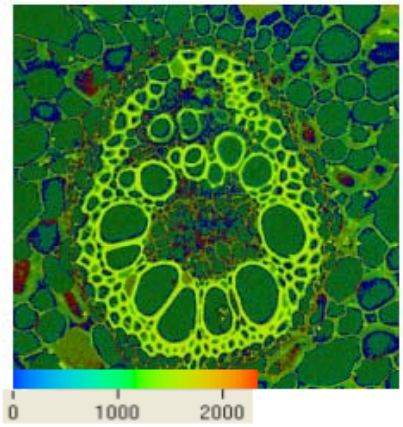
Fluorescence life time imaging microscopy (FLIM)



Conventional confocal intensity image

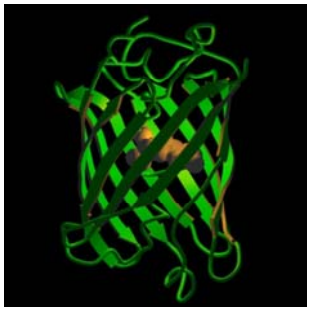


Fluorescence lifetime image /ps

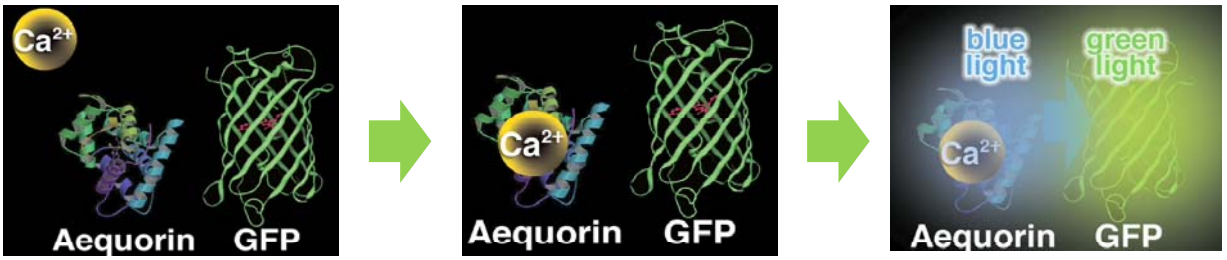
What is Green Fluorescent Protein (GFP)?



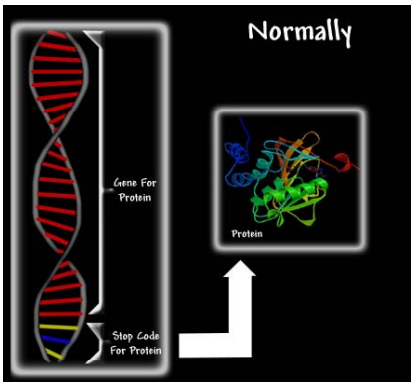
Aequorea victoria jellyfish

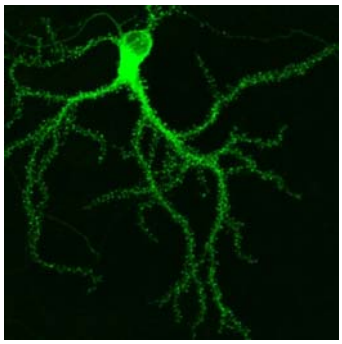


GFP

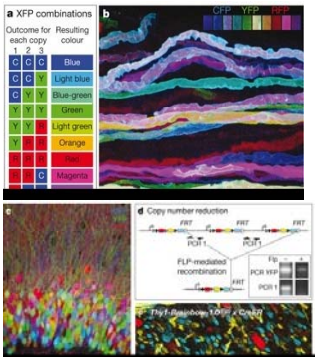


Applications of GFP

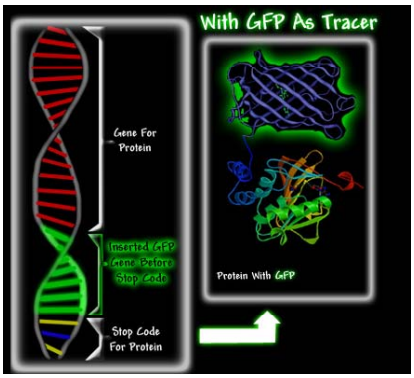




Neuron GFP

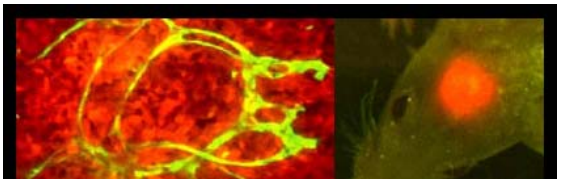


Livet et al., Nature **450**, 56 (2007)





GFP cloned 쥐와 보통 쥐



암세포들 (붉은색)과 혈관 (녹색)

http://www.conncoll.edu/ccacad/zimmer/GFP-ww/GFP-1.htm

The Nobel Prize in Chemistry 2008

"for the discovery and development of the green fluorescent protein, GFP"

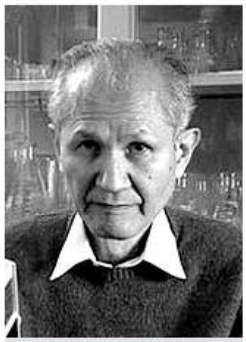


Photo: J. Henriksson/SCANPIX

Osamu Shimomura

1/3 of the prize

USA

Marine Biological Laboratory (MBL)
Woods Hole, MA, USA

b. 1928



Photo: J. Henriksson/SCANPIX

Martin Chalfie

1/3 of the prize

USA

Columbia University
New York, NY, USA

b. 1947



Photo: UCSD

Roger Y. Tsien

1/3 of the prize

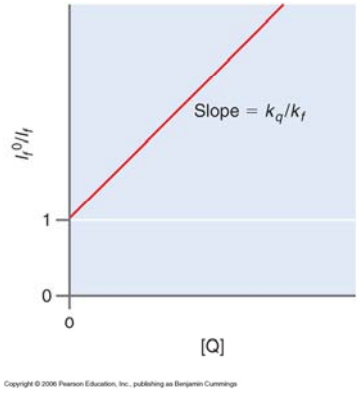
USA

University of California
San Diego, CA, USA

b. 1952

형광-소광 반응과 Stern-Volmer 식

- 소광 (quenching): 분자 간 충돌에 의한 비활성화, 에너지 전달 또는 전자전달 등에 의해 전자의 들뜬 상태의 수명이 줄어드는 것. 소광에 의해 형광 양자수득률이 감소한다.
- Stern-Volmer 식
  - 연속 광원에 의한 빛의 흡수: $S_0 + h\nu \xrightarrow{k_a} S_1$
  - 형광 반응: $S_1 \xrightarrow{k_f} S_0 + h\nu$
  - 소광 반응: $S_1 + Q \xrightarrow{k_q} S_0 + Q$



$$\frac{d[S_1]}{dt} = k_a[S_0] - k_f[S_1] - k_q[Q][S_1] \approx 0$$

$$\Rightarrow [S_1] = \frac{k_a[S_0]}{k_f + k_q[Q]}$$

$$I_f = k_f[S_1] = \frac{k_f k_a[S_0]}{k_f + k_q[Q]} \Rightarrow \frac{1}{I_f} = \frac{k_f + k_q[Q]}{k_f k_a[S_0]} = \frac{1}{k_f k_a[S_0]}\left(1 + \frac{k_q}{k_f}[Q]\right)$$

$$\frac{I_f^0}{I_f} = 1 + \frac{k_q}{k_f}[Q] \qquad (I_f^0 \equiv k_a[S_0])$$

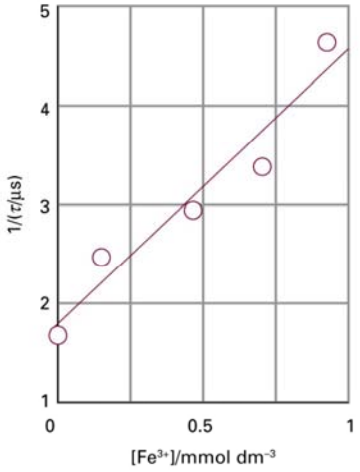
형광-소광 반응과 형광 수명 측정

- 펄스 광원에 의한 k_f 와 k_q 결정 방법

$$\frac{d[S_1]}{dt} = -k_f[S_1] - k_q[Q][S_1] = -(k_f + k_q[Q])[S_1]$$

$$\Rightarrow [S_1] = [S_1]_0 e^{-t/\tau} \qquad \left(\tau = \frac{1}{k_f + k_q[Q]}\right)$$

$$\tau = \frac{1}{k_f + k_q[Q]} \Rightarrow \frac{1}{\tau} = k_f + k_q[Q]$$





(예제) 수용액에서 용해된 O_2에 의한 트리토판 형광의 소광 반응을 348 nm에서 방출되는 형광 수명을 측정함으로서 관측하였다. 얻어진 자료가 다음과 같을 때 소광 속도상수를 구하여라.

$[O_2]$ / (10^{-2} mol dm^{-3})	τ/(10^{-9} s)
0	2.6
2.3	1.5
5.5	0.92
8.0	0.71
10.8	0.57

